EE3111 Lab 2: Exploratory Data Analysis & Distribution Fitting
In-class Live Coding Challenge

Department of Electrical Engineering, IIT Madras

Demo Date: Feb 11, 2026 during class

1 Part A: Maternal Smoking (EDA)

A few other ways to visualize the data:

1. Plot the two empirical CDFs, one of smokers and one of non-smokers, in the same plot.
2. Create a stripplot for the data with the same axes as the box plot.
3. Create a violin plot (with the same axes as the box plot), which is often considered to be more informative than the box-plot.

2 Part B: Video Game Analytics (Distribution Fitting)

We consider various metrics to verify the goodness of fit of the MLE and MOM methods.

1. Compute the log likelihood function (denoted ℓ) evaluated at the MLE estimate $(\hat{\alpha}_{\text{MLE}}, \hat{\beta}_{\text{MLE}})$ and the Method of Moments (MOM) estimate $(\hat{\alpha}_{\text{MOM}}, \hat{\beta}_{\text{MOM}})$. Verify (up to small numerical errors) that

$$\ell(\hat{\alpha}_{\text{MLE}}, \hat{\beta}_{\text{MLE}}) \geq \ell(\hat{\alpha}_{\text{MOM}}, \hat{\beta}_{\text{MOM}}).$$

2. Select several arbitrary values of (α, β) and compute the corresponding log-likelihood values.

   Verify numerically for those choices that

$$\ell(\hat{\alpha}_{\text{MLE}}, \hat{\beta}_{\text{MLE}}) \geq \ell(\alpha, \beta)$$

3. The Kolmogorov–Smirnov statistic measures the maximum absolute difference between the empirical cumulative distribution function (ECDF) and the fitted model CDF:

$$D(\alpha, \beta) = \sup_x |F_n(x) - F(x \mid \alpha, \beta)|,$$

where $F_n(x)$ is the empirical CDF and $F(x \mid \alpha, \beta)$ is the Gamma CDF.

Compare the KS values of the MOM estimate and the MLE estimate. Also pick several arbitrary values of α and β and compute the KS values. You can use the `scipy.stats.kstest` for this.

Remark. While the MLE estimate maximizes the log likelihood function, it does not necessarily minimize the KS value.